

Highlights

Surface Tension Induced Climbing Kinematic Robot (S.T.I.C.K.R.)

Erik Goeke, Ryan Cohen

- Demonstrate a small robot that can drive on a horizontal surface in microgravity using the surface tension of a fluid.
- Demonstrate that a fluid with a high surface tension can cause an object to stick to the wetted surface.

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ABSTRACT

This report covers the design, development, and testing of STICKR, a wall crawling robot in microgravity. Microgravity in space eliminates the normal force of gravity experienced on planets, and it renders traditional wheeled methods for robotic control less effective. STICKR explores the use of ferrofluids, and their unique properties of both magnetism and surface tension. STICKR was developed and designed with a focus on a lightweight and simple construction, and also seeks to provide valuable data for use in other ferrous fluid microgravity adhesion experiments.

1. Introduction

The development of a robotics system capable of operating in microgravity is critical in the growing space domain. Applications of robotic systems in space include in-orbit inspections and repairs of spacecraft, as well as deep space exploration. Space vehicles do not experience the same gravity as on Earth, limiting the locomotion of in-space robotics vehicles, and rendering wheels and legs near useless. As a result, new methods of mobility must be designed and explored in order for robotic vehicles to operate effectively in space.

Currently there are methods for robotic space vehicles which make the use of thrusters and reaction wheels, such as Astrobe: National Aeronautics and Space Administration (2025). These methods are effective, however they face problems with power consumption and increased mechanical complexity. Adhesion based solutions present an interesting alternative, but these designs face control challenges - namely attaching and detaching from surfaces. Magnets are a promising design choice, however it cannot be guaranteed that every surface of a spacecraft will be magnetic.

This project investigates the use of fluids as a mechanism for surface tension in order to keep a small and light robot attached to any surface in microgravity. Specifically, using a ferrous fluid attached to the robot via magnets. Ferrous fluids (or ferrofluids) are defined as "a suspension of magnetic particles in a carrier liquid, with high saturation magnetization" Elsevier (2026). In other words, they are fluids which respond to a magnetic field, allowing them to leverage both magnetic and surface tension forces. NASA used a similar method when using ferrofluids to manage liquid rocket fuel by attracting them to a fuel tank Elsevier (2026).

The Surface Tension Induced Climbing Kinematic Robot, or STICKR, explores the aforementioned properties of ferrofluids and seeks to validate their applicability to space

missions. The goal of the project is to create a robot capable of driving on a horizontal surface in microgravity in a controlled manner. This report will cover the design, manufacturing, and experimental testing of the STICKR system, and will also discuss technical challenges and opportunities for future development.

In order to test the abilities of the STICKR robot, it will fly aboard a parabolic zero-gravity flight, which are used to simulate microgravity environments. During the flight, the aircraft flies in a parabolic trajectory, where at the peak, the engines are shut off, and the plane will be in a state of free fall for about 15 seconds. In this time the entire cabin will be in a state of microgravity. Zero Gravity Corporation (2026) An example of these parabola can be seen in Figure 1.

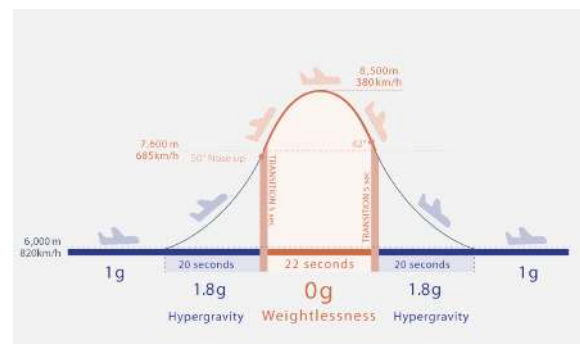


Figure 1: An example parabola for a microgravity flight

2. Science/Technology Objectives and L-1 Requirements

The primary science and technology objectives for STICKR are listed below.

Objective 1: Demonstrate a small robot capable of locomotion in microgravity using surface tension-based adhesion.

Objective 2: Demonstrate that ferrofluid can generate sufficient adhesion to enable controlled attachment to a wetted surface.

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Table 1: L-1 Requirements for STICKR

L-1 ID	Requirement Description
STICKR-F-1.0	The system shall demonstrate controlled movement using ferrofluid-based adhesion in a microgravity environment.
STICKR-P-2.0	The system shall sustain continuous operation for a minimum mission duration.
STICKR-I-3.0	The system shall interface with metallic surfaces without permanent modification.
STICKR-E-4.0	The system shall prevent ferrofluid leakage and environmental contamination.
STICKR-PH-5.0	The system shall comply with the 0.5U pocket payload mass constraint.
STICKR-O-6.0	The system shall operate via Arduino Nano control without active ferrofluid regulation.
STICKR-D-7.0	The system shall use ferrofluid-based adhesion compatible with metallic surfaces in microgravity conditions.

Table 2: Mission Success Criteria for STICKR

MSC ID	Success Criteria (Minimal / Meets / Exceeds)
MSC-1	Adhesion: >30 s (min), >1 m traversal (meets), 90° corner (exceeds)
MSC-1A	Load: own weight (min), 1.5× (meets), 2× (exceeds)
MSC-1B	Acceleration: >0.02g (min), >0.05g (meets), >0.2g (exceeds)
MSC-2	Retention: <0.3 mL loss (min), no visible loss (meets), none measurable (exceeds)
MSC-2A	Contamination: no leakage (min), no residue (meets), none measurable (exceeds)
MSC-2B	Retention under accel.: >0.02g (min), >0.05g (meets), >0.1g (exceeds)
MSC-3	Mobility: single-axis (min), two-axis (meets), autonomous (exceeds)
MSC-3A	Endurance: >1 m (min), >2 m (meets), extended (exceeds)
MSC-3B	Accuracy: >0.1 m dev. (min), <0.05 m (meets), none visible (exceeds)
MSC-4	Duration: >5 min (min), >10 min (meets), stable extended (exceeds)
MSC-4A	Repeatability: >2 runs (min), >3 runs (meets), >5 runs (exceeds)

The L1 requirements and the mission success criteria, which drove the design of STICKR, can be found in tables 1 and 2. These requirements were developed from the science objectives mentioned previously in this report. The L1 requirements were all set as subgoals used in the development towards completing main science objectives. In other words, in order to achieve the both objectives 1 and 2, the weight and size requirements need to be met. The L1 requirements are also set as a design plan in order to achieve the main science objectives. The mission success criteria follows a similar process. In order to determine whether or not the mission was a success, is dependent upon its completion of the main objectives. In order to validate the completion of these science objectives, criteria is set for individual parts of what makes the experiment a success.

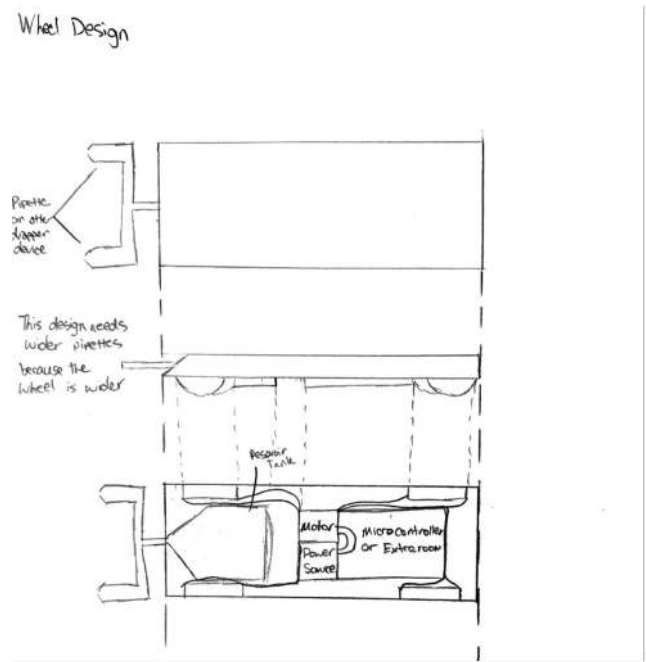
3. Experimental Design

The STICKR experiment is designed to test the feasibility of adhesion using the surface tension of a fluid in a microgravity environment. The experiment is set up through the design and manufacturing of a small, lightweight robot, to be tested in microgravity.

3.1. Initial Design

The initial design of STICKR explored a variety of options for locomotion and the methods of using fluids. A tank drive tread design was selected as more viable than robotic legs, due to the increased fluid contact perimeter allowed by a larger contact surface. Additionally, designing a driving mechanism for a walking robot introduces significant complexities, and therefore additional mass, which is a quantity to be minimized for this project.

Initially, a pipette dropper was considered as a method of placing the fluid in front of the robot, providing a layer of fluid to drive across. Several fluids were considered for this, including silicone oil, chosen for its high surface tension value. This concept was replaced with the simpler method of attaching magnets to the treads and placing a drop of a ferrous fluid on each magnet. Therefore the fluid remains on the robot for the duration of the mission, and no additional residue is left behind when driving. The initial concept for a pipette dropper design is shown in Figure 2

**Figure 2: Initial Proposed STICKR Design**

This design evolved into what can be seen in figure 3. The second iteration of the design included a servo motor to drive the treads with a

1:2 gear ratio to slow the rotational speed and provide additional torque for driving. The treads include circular magnets evenly spaced across, which were later replaced with bar magnets, providing additional contact perimeter.

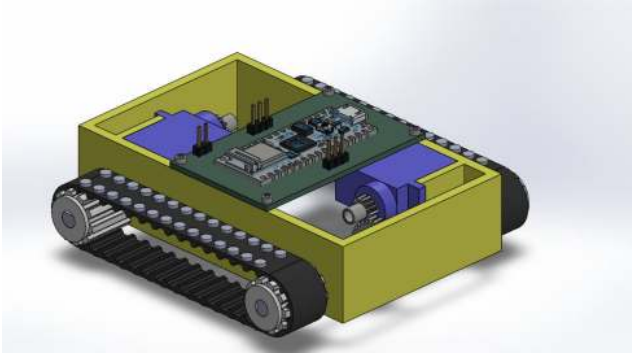


Figure 3: Initial CAD design of the proposed idea for the STICKR system

The final design, shown in figure 4, removed one of the two servo motors to minimize weight, and replaced the circular magnets with bar magnets.

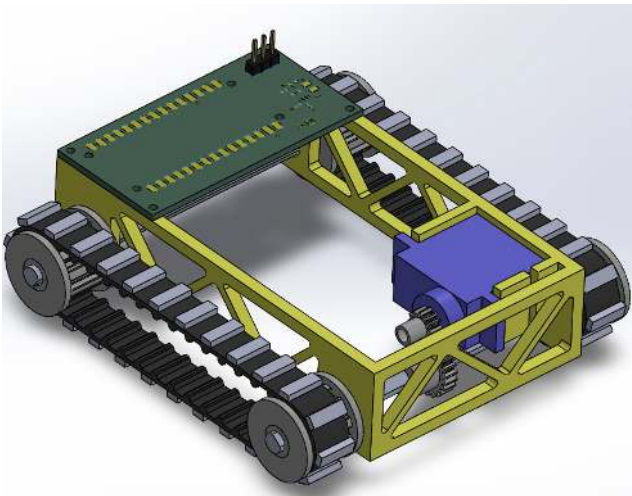


Figure 4: Final CAD design

3.2. Electronics

The electronic systems used to drive and control STICKR were designed to run as simply and efficiently as possible. The microcontroller on-board is an Arduino Nano 33 BLE Sense, chosen for its small form factor and low mass. The Arduino was integrated into a custom PCB where a 9V input signal can power the controller. The Arduino drives the servo with a PWM signal from one of its digital I/O pins. Also included on the PCB is a DROK buck converter which steps the 9V input into a 5V power signal for the servo. See the PCB in figure 5.

One requirement for the STICKR robot is to include an accelerometer on board which can measure the disturbance forces in the environment, such that if the robot fails to adhere, it can be determined in what direction the disturbance acted. Integrated into the Arduino microcontroller is a LSM9DS1 inertial measurement unit (IMU). During STICKR's operation, the IMU collects acceleration data at a rate of 10 Hz, and stores the accelerations in the three primary axes to the built-in RAM of the Arduino. At the end of operation, this RAM can be offloaded via the serial port and saved for further analysis.

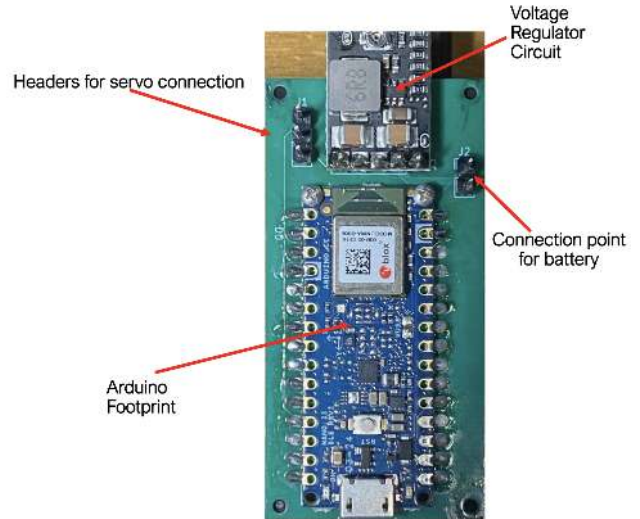


Figure 5: STICKR PCB

To reduce the mass of STICKR, the 9 volt power input is separate from the robot and must be set by the operator. For this particular experiment, the input comes from six AA batteries which are harnessed to the PCB via thin copper wires, which will reduce the overall disturbance force on the robot while the batteries are held nearby. Thicker wires would introduce additional forces, particularly in microgravity, that may disturb the experiment. See figure 6, where a traditional COTS battery clip connector breaks out into very thin wires.



Figure 6: Thin harnessing used for the battery connection

3.3. Bill of Materials

Table 3 shows the bill of materials for the final design to document the overall cost of the STICKR robot. There were some budgeting constraints for this project, in addition to some shipping delays which compressed the schedule for assembly/testing. Table 4 shows the mass budget, and the allocation of STICKR's final mass.

Table 3: Bill of Materials

Item	Unit Cost	Qty	Total Cost
Carbon Fiber/Resin (3D Print)	\$3.00 (est.)	—	\$3.00
Servo	\$7.50	1	\$7.50
Arduino	\$23.10	1	\$23.10
Tread	\$5.62	2	\$11.24
PCB	\$40.00	1	\$40.00
Magnets	\$0.26	26	\$6.76
Glue	\$12.45	1	\$12.45
Battery holder	\$5.99	1	\$5.99
Battery connector	\$0.56	1	\$0.56
Total Cost:			\$110.60

Table 4 includes the final mass budget for the STICKR robot, with the overall estimated mass coming to 58.7 grams. This is well within the limits which had initially estimated 100 grams for the robot.

Table 4: Mass Budget

Part	Unit Mass (g)	Qty	Total Mass (g)
Structure (Carbon Fiber / 26 Tough 2000 3D Print)		1	26.00
Treads (Urethane Rubber)	3.00	2	6.00
PCB + Arduino + Headers	7.5	1	7.50
Servo	10.1	1	10.10
Magnets (Neodymium)	0.35	26	9.10
Total Mass:			58.70

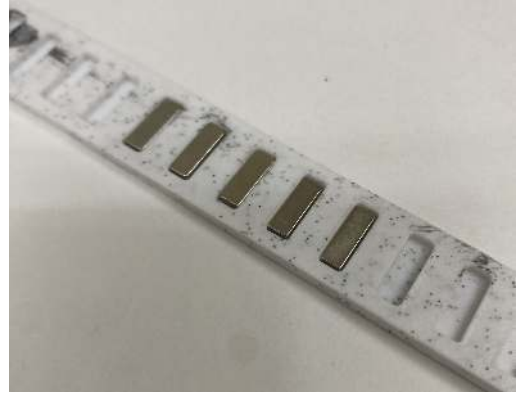
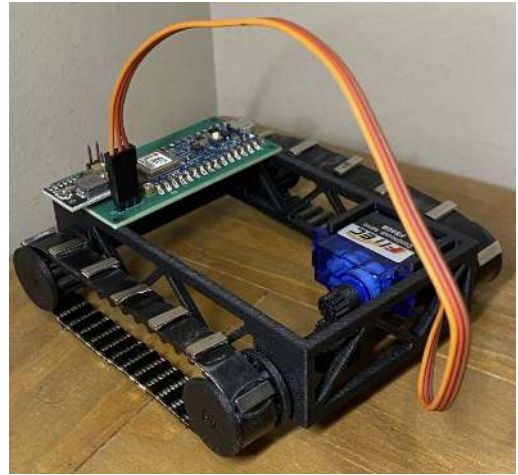
3.4. Assembly

The main focus in the manufacturing process was to keep the design compact and lightweight, while also keeping the system simple and easily repeatable. This meant the manufacturing methods and parts used were very common and accessible.

- The chassis, axles, and pulleys were all printed on a Bambu Labs A1 printer in carbon fiber to keep the weight down.
- The storage case was also printed on the Bambu Labs printer in PLA.
- The servo gears were printed in resin on a Formlabs SLS printer

One major challenge that presented itself during the manufacturing process was applying the magnets onto the rubber tread. A particular flexible super-glue was used to ensure that the magnets remain on the treads as they rotate around the edges. The initial calculations suggested that the

magnets should be spaced 4 mm apart to provide enough holding force, however this proved difficult to assemble by hand in practice. A support structure was designed to accurately position each magnet, however the structure itself would stick to the treads when glued. For future use, some form of thin plastic or a layer of oil could be used between the glue and the support structure to prevent this from happening. For the final assembly, this spacing distance was doubled, with 13 magnets stuck to each tread. The support structure is shown in figure 7.

**Figure 7: Magnet Assembly Support Structure****Figure 8: STICKR Robot**

After assembly was complete (see figure 8) The 3D printed gears created another problem. Like the rest of the structure, the gears were printed with carbon fiber on a Bambu Labs A1 printer. The teeth on the gears weren't printed with enough resolution to mesh properly, leading to the robot being unable to drive. For this reason, STICKR was updated to have its gears reprinted on a Formlabs resin printer, which was able to create gears with stronger teeth that mesh well. See figure 9, note the pulleys on one axle were also replaced with resin printed parts due to availability for the reprinted version.

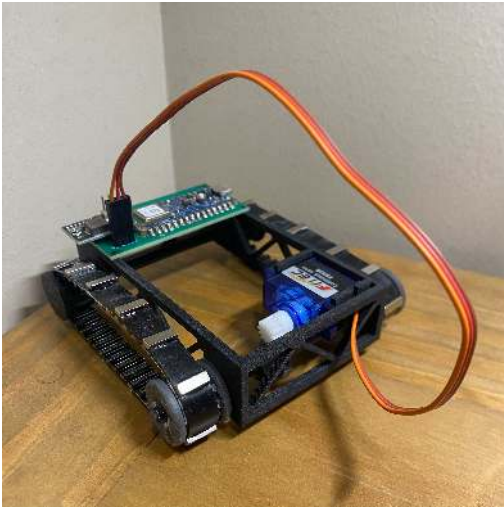


Figure 9: Final STICKR robot with resin printed gears

The final measurements of the system are as follows

- **Mass:** 57.0 grams
- **Volume:** $100 \times 95 \times 40$ mm
- **Power:** 9V input, ~ 160 mA max draw
 - ~ 1.44 W for 15 sec
- **Data:** 4.86 KB .csv file per run
- **Holding Force (est.):** 8.62 mN
- **Cost:** \$110.60

3.5. Surface Tension Calculations

The following is the background math that was used to determine the surface tension holding force of the STICKR robot.

To adhere to a surface, the robot's surface tension holding force must be greater than the disturbing accelerations inside the microgravity parabolic flight. The flight is known to reach accelerations of 0.01 G, or 0.0981 m/s^2 . Given the mass of the robot, the disturbance forces F_d on STICKR are calculated using:

$$F_d = m \times a_d = 0.057 \times 0.0981 = 0.0056 \text{ N} = 5.6 \text{ mN} \quad (1)$$

The surface tension force F_γ between a drop of fluid and a surface is given by

$$F_\gamma = \gamma \times P \times \sin \theta \quad (2)$$

Where γ is the surface tension of the fluid (a material property), P is the perimeter of the contact surface, and θ is the "wetting angle" which is a unique property of the fluid and the surface. For the ferrofluid being used, the surface tension is estimated to be 60 mN/m with a contact angle of 45° . The main variable to be controlled is the contact perimeter. The robot uses bar magnets with a contact surface area of $3/8$ by $1/8$ inches, giving a perimeter of 1 inch per magnet. Based on the tread geometry, with 13 magnets per tread, there are four magnets per tread in contact with the surface at any given moment. Doubled (because STICKR has two treads) gives a total contact perimeter of 8 inches, or 20.32 cm. Completing the calculation, this gives a holding force of 8.62 mN. This exceeds a safety factor of 1.5, and shows that the surface tension holding force should be capable of counteracting the disturbance forces. This safety factor could be improved more closely spaced magnets.

4. Testing

The final test of this robot will be onboard the parabolic flight with ferrofluid on each of the magnets to test the surface tension adhesion. Prior to the microgravity flight, the robot is to be tested in full gravity to ensure

the drive mechanism is stable. As an additional stress test, STICKR can be placed on a vertical magnetic surface to drive up a wall in full gravity, however overcoming the magnetic force to drive would require more torque than STICKR would need in practice to overcome the surface tension force. Figures 10 and 11 depict these tests.

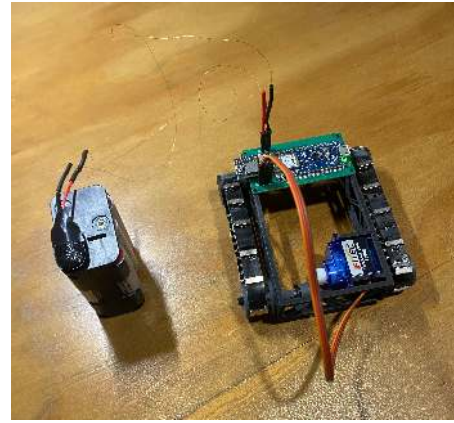


Figure 10: Test on a horizontal nonmagnetic surface

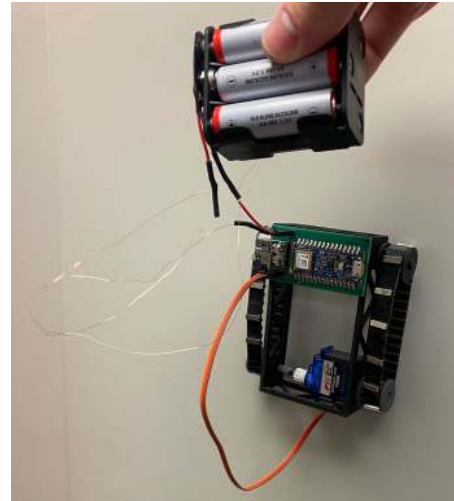


Figure 11: Test on a vertical magnetic surface

The full steps for experiment operation are as follows:

1. Plug in 9V power supply
2. 5 second delay – position robot on surface
3. Allow robot to drive for 15 seconds
4. Leaving battery plugged in, connect to laptop via micro-USB port
5. Open serial monitor (i.e. PuTTY)
 - Press 'D' on the keyboard to dump output
6. Save as .csv file (automatic if using PuTTY)

An example of the expected output is shown in Figure 12

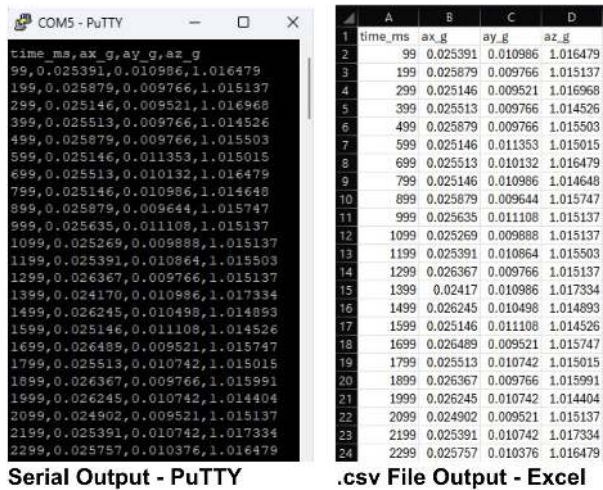


Figure 12: Expected Putty and Arduino output

5. Next Steps

Future development of the STICKR system will be to test it on a Zero-G flight to measure its abilities in microgravity. These tests will be compared to both the mission success criteria, as well as the science requirements. From this point, the system will be improved based on the results of the microgravity flight.

6. Acknowledgments

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Erik is an Aerospace Engineering Masters student at the Georgia Institute of Technology. He does research in robotics, as well as AI and machine learning. Robotics has been a central part of his research for three years, and he seeks to apply his knowledge to all types of space robotics missions.



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